STAGE 1 TECHNICAL REPORT

Garud FX: Unmanned Aerial Vehicle Forensic Toolkit

PUSHPAK Grand Challenge 2026 (Objective 1: Drone Forensics Toolkit Development)
National Mission on Drone Technology | Ministry of Electronics and Information Technology (MeitY), Govt. of India

TEAM AND SUBMISSION INFORMATION

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1. EXECUTIVE SUMMARY

1.1 Solution Vision & Legal Standards Compliance

Unmanned Aerial Vehicles (UAVs) have evolved into critical autonomous infrastructure across commercial, civil, and military operations. However, their widespread availability has introduced grave security vulnerabilities, including unauthorized airspace incursions, critical infrastructure surveillance, trans-border contraband delivery, and cyber-physical interference. In forensic investigations and judicial proceedings, the integrity, mathematical repeatability, and physical provenance of extracted UAV flight data represent the foundational pillars of courtroom admissibility.

To address this mandate under the PUSHPAK Grand Challenge 2026, Team Halos presents **Garud FX: Unmanned Aerial Vehicle Forensic Toolkit**—an indigenous, multi-vendor, hardware-integrated forensic software suite engineered for end-to-end UAV investigation. Garud FX strictly adheres to international digital evidence standards **ISO/IEC 27037:2012** (Identification, collection, acquisition, and preservation of digital evidence) and **NIST SP 800-86** (Guide to Integrating Forensic Techniques into Incident Response). Furthermore, Garud FX natively incorporates statutory compliance workflows for **Section 63 of the Bharatiya Sakshya Adhiniyam (BSA), 2023** (replacing Section 65B of the Indian Evidence Act, 1872). Specifically, Garud FX auto-populates a Section 63(4)-compliant digital evidence certificate template, pre-filled with dual cryptographic hash manifests, binary offsets, and hardware device signatures, ready for statutory execution by the designated device custodian and independent expert examiner.

1.2 Key Architectural Innovations

- **Hex-Offset Provenance Mapping:** Every extracted telemetry event (geodetic coordinates, IMU dynamics, battery voltage, flight mode) maintains an immutable, bi-directional link to its exact raw hexadecimal byte offset within the underlying binary evidence dump (e.g., Offset 0x0001F820). Forensic examiners can scrub between 3D spatial visualization and raw hex inspector view for zero-trust court verification.
- **Air-Gapped 3D Offline Trajectory Reconstruction:** Garud FX embeds a self-contained local MBTiles raster/vector mapping server and an offline 3D spatial trajectory engine. Examiners perform complete 3D flight path analysis, altitude profiling, and camera orientation visualization inside air-gapped cleanrooms without internet connectivity.
- **Signature-Matching & Fallback Binary Carving:** UAV crash incidents frequently trigger sudden power disconnects, corrupting log files missing end-of-file (EOF) structures. Garud FX pattern-matches raw byte streams against 9 pre-registered multi-vendor format signatures with a fallback binary carving engine to recover intact telemetry up to the microsecond of power loss.
- **In-Stream Dual Cryptographic Hashing & SQLite Case Packaging:** During the initial hardware read pass, Garud FX computes simultaneous SHA-256 and BLAKE3 cryptographic hashes without secondary I/O overhead. All custody actions are recorded in an append-only hash-chained custody ledger (custody.ledger) and packaged into a single, queryable SQLite forensic database (.garud_db format).

> **■ EXECUTIVE SUMMARY HIGHLIGHT**
> *Garud FX unifies physical bit-stream acquisition, low-level binary carving, air-gapped 3D spatial analytics, raw hex-offset provenance, SQLite case packaging (.garud_db), and Section 63(4) BSA legal certificate template generation into a single workstation suite.*

1.3 Target Platforms & Evidence Taxonomy

- **Target UAV Autopilots:** DJI (Mavic 3, Phantom 4, Inspire 2, Matrice 300 RTK, Mini series), PX4 Autopilot, ArduPilot (APM / Pixhawk / Cube), Betaflight Blackbox, Autel Explorer, and Yuneec DataPilot.
- **Physical Evidence Sources:** Physical raw bit-stream dumps (.dd / .raw) of onboard eMMC flash chips, MicroSD cards, and flight controller flash memory.
- **Logical Evidence Sources:** Android Debug Bridge (ADB) extractions from companion Ground Control Station (GCS) mobile apps (DJI Fly, DJI Go 4, DJI Pilot 2, Autel Explorer, Yuneec DataPilot).
- **Hardware Communications:** Direct serial USB UART interfaces implementing MAVLink v2.0 micro-chunk extraction protocols.
- **Multimedia & Payload Evidence:** Video SRT subtitle telemetry streams and JPEG EXIF geotagged metadata extracted from camera payloads.

1.4 Current Development Status: Functional Prototype

Garud FX is currently a fully functional software prototype operating with modular Python extraction engines, binary parsers for 9 independent UAV log formats, an analytical correlation engine, an air-gapped Web GUI (Leaflet 2D, Cesium 3D, and 3-column raw hex inspector), and an automated PDF courtroom report generator. Comprehensive testing has validated Garud FX's capability to carve corrupt flight logs, detect GPS spoofing, enforce hash-chained custody ledgers, package queryable .garud_db SQLite cases, and auto-populate BSA Section 63(4) certificates.

2. UNDERSTANDING OF THE PROBLEM

2.1 Technical Challenges in Drone Digital Forensics

- **Volatile RAM Buffers & Abrupt Crash Disconnects:** Flight controllers buffer high-frequency telemetry in volatile SRAM. Mid-air collisions or battery ejections cut main power rails instantly, preventing clean OS unmounts or EOF marker writes. Standard parsers fail on such truncated binary files.
- **Multi-Vendor Proprietary Encrypted & Obfuscated Schemas:** UAV manufacturers employ closed binary formats. DJI utilizes XOR obfuscation and AES-128 payload encryption across DAT file versions (Fly0182, P3/P4, Mavic 3). Open-source platforms (PX4, ArduPilot) utilize dynamic binary structure packing requiring runtime schema extraction.
- **OS-Level Metadata Contamination:** Connecting recovered UAV media directly to host workstations (Windows/macOS) risks immediate evidence spoliation due to automated hidden system file writes (.DS_Store, System Volume Info) and modified access timestamps.
- **Timezone Skew & Multi-Clock Drift:** Drone investigations involve multiple time domains: UTC timestamps from onboard GPS receivers, microsecond epoch counters from flight controller clocks, and local time settings on operator mobile GCS devices.
- **Air-Gap Cleanroom Requirements:** Secure forensic cleanrooms operate under complete network isolation. Traditional forensic software relying on online mapping APIs (Google Maps, Mapbox) fail in air-gapped environments.

2.2 Target Forensic Investigation Scenarios

- **Airspace Intrusion & Critical Infrastructure Defense:** Reconstructing flight paths of rogue UAVs over restricted airspace, airports, or military installations to pinpoint launch coordinates and operator positioning.
- **Contraband Delivery & Border Smuggling:** Analyzing recovered crash or intercepted drones used for smuggling narcotics or contraband across international borders or penal perimeters.
- **Mid-Air Collision & Aircraft Incident Reconstruction:** Investigating aerial collisions with commercial aircraft or infrastructure to determine pilot inputs, battery collapse, or mechanical failure.
- **GPS Spoofing & Cyber-Physical Attack Verification:** Detecting electronic warfare interference, GPS spoofing, or command injection targeted at autonomous drone platforms.

2.3 Evidence Taxonomy & Industry Limitations Solved

Garud FX categorizes digital evidence into two distinct operational tiers:
1. Onboard Blackbox Telemetry: High-frequency (10Hz - 250Hz) physical dynamics (IMU accelerometers, gyroscopes, barometric pressure, motor current, battery cell voltages, raw GPS PVT, flight state machines) and UAV configuration and parameter files (e.g., ArduPilot .param files, PX4 parameter trees, fail-safe threshold registries, PID tuning profiles).
2. Ground Control Station (GCS) Telemetry: Logical application data on mobile controllers (waypoint XMLs, joystick inputs, operator mobile GPS establishing pilot ground truth, app crash

logs, companion app config files).

Existing commercial forensic tools treat UAV storage as standard mobile endpoints or generic mass storage, failing to parse proprietary flight dynamics or provide spatial 3D kinematics. Garud FX bridges these gaps through native multi-vendor binary parsers, offline 3D spatial mapping, raw hex-offset provenance, and statutory BSA Section 63(4) certificate template generation.

3. PROPOSED DRONE FORENSIC TOOLKIT

3.1 Garud FX System Architecture

Garud FX is constructed around a 5-layer modular software architecture designed for high scalability, air-gapped security, and complete vendor independence:

• Layer 1 (Acquisition Layer): Interfaces with raw physical media (.dd eMMC/SD), Android GCS devices via ADB, and live serial USB MAVLink hardware ports using non-destructive streaming.

• Layer 2 (Preservation & Custody Layer): Computes simultaneous SHA-256 and BLAKE3 hashes, maintains an append-only hash-chained custody ledger (custody.ledger), compiles artifacts into an indexed SQLite database (.garud_db format via GarudCasePackager), and manages Section 63(4) BSA certificate templates.

• Layer 3 (Parsing & Carving Layer): Houses 9 multi-vendor binary parsers (PX4 ULog, ArduPilot DataFlash, DJI DAT, DJI CSV, Autel, Yuneec, Betaflight, MAVLink, EXIF/SRT) equipped with pattern-matching carving fallback engines.

• Layer 4 (Analytics & Correlation Engine): Executes spatial-temporal synchronization, Haversine metrics, geofence ray-casting, velocity anomaly detection, and GPS spoofing verification.

• Layer 5 (Presentation & Reporting Layer): Delivers an air-gapped Web GUI (Leaflet 2D, Cesium 3D globe, 3-column raw hex inspector linked to .garud_db queries) and an automated PDF courtroom exhibit generator.

3.2 Supported UAV Platforms & Technical Format Matrix

Platform	Format	Acquisition	Parsed Parameters	Integrity & Carving
PX4 Autopilot	.ulg (ULog Binary)	Serial USB / Physical SD	Geodetic (Lat/Lon/Alt), IMU, Battery V/A, Attitude, Parameter Trees	SHA256 + BLAKE3 + Carving
ArduPilot (APM)	.bin (DataFlash), .log	Serial USB / Physical SD	GPS, POS, ATT, BAT, RCIN/RCOUT, MODE, .param Config Files	Dual Hashing + Header Sync
DJI Onboard	.DAT (P3/P4/Mavic)	ADB / Physical SD	GPS, IMU, Motor Current, Battery Temp, RC Channels, Configs	Public XOR / Header Parsing (AES-128 via MS-02)
DJI Flight Record	.csv / Encrypted .txt	ADB Mobile Pull	OSD Coordinates, Battery %, Joysticks, App Warnings	Header Byte Classification

Autel Explorer	.csv (Flight Logs)	ADB Mobile Pull	GPS, Altitude, Speed, Battery, RC Deflection, App Configs	In-Stream Hashing
Yuneec DataPilot	.csv (Flight Logs)	ADB Mobile Pull	Lat/Lon, Alt, Compass, Voltage, Mode State, System Settings	In-Stream Hashing
Betaflight	.bbl (Blackbox)	Physical SD / Micro Flash	High-Freq Gyro, Motor PWM, PID Loop, Battery, Rates Config	Magic Signature Carving
MAVLink Telemetry	.tlog (Telemetry Log)	Serial USB Hardware	RAW_IMU, GPS_RAW_INT, ATTITUDE, PARAM_VALUE Configs	Micro-Chunk Validation
Media EXIF / SRT	.JPG, .MP4, .SRT	Payload SD Card	Sub-second Geotags, Camera Pitch/Roll, ISO, Shutter	Metadata Extraction

3.3 Digital Evidence Acquisition Methodology

Garud FX mandates non-destructive physical and logical acquisition workflows conforming to ISO/IEC 27037 Clause 6.3:
1. Chunked Read-Only Physical Imaging: Block-level chunked reads (64KB chunks) wrapped in read-only file streams (rb). Integrates with physical USB write-blockers and queries OS mount controllers to suppress write-backs.
2. Hardware Serial MAVLink Micro-Chunk Extraction: Communicates over serial USB using MAVLink v2.0 protocol. Transmits LOG_REQUEST_LIST and fetches log binary data via LOG_REQUEST_DATA in 240-byte micro-chunks with 16-bit CRC checksum validation.
3. Logical Android ADB GCS Extraction: Interfaces via adb to pull companion GCS application logs (DJI Fly, DJI Go 4, Autel Explorer) directly from mobile partitions.

3.4 Evidence Preservation & Cryptographic Integrity

Evidence integrity is maintained through in-stream dual cryptographic hashing (SHA-256 and BLAKE3) calculated during the initial raw byte read. Actions are logged in an append-only JSONL hash-chained custody ledger (custody.ledger) containing monotonic timestamps, action types, investigator signatures, and cryptographic parent-hash linkage. The complete case case is packaged into an indexed SQLite forensic database (.garud_db) for tamper-evident distribution.

To satisfy Indian statutory requirements under Section 63 of the Bharatiya Sakshya Adhiniyam (BSA), 2023, Garud FX auto-populates a Section 63(4)-compliant digital evidence certificate template, pre-filled with SHA-256/BLAKE3 hash manifests, binary offsets, and hardware device signatures, ready for statutory execution by the designated device custodian and independent expert examiner.

3.5 Data Extraction & Signature-Matching Carving Engine

To salvage truncated crash logs, Garud FX pattern-matches raw byte streams against 9 pre-registered multi-vendor format signatures (0x55 0x4C 0x6F 0x67 ULog, 0xA3 0x95 DataFlash, 0x46 0x4C 0x59 DJI DAT) with a fallback binary carving engine to unpack intact telemetry packets up to the corruption horizon. All data is normalized into a canonical time-series schema (timestamp_utc, lat, lon, alt_msl, roll, pitch, yaw, battery_v, flight_mode, raw_hex_offset).

4. DATA ANALYSIS & AUTOMATED FORENSIC REPORTING

4.1 Multi-Source Timeline Synchronization

The analytical correlation engine (analytics/correlation.py) aligns disparate time-series evidence across onboard blackbox logs, mobile GCS databases, and controller logs. The module establishes UTC GPS reference anchors to calculate microsecond clock drift, interpolates missing packets, and merges multi-source telemetry into a synchronized event timeline.

4.2 Kinematic Anomaly & Cyber Threat Detection

- **Stationary Hovering vs. Active Flight:** Analyzes horizontal speed vectors (< 0.5 m/s over > 30s) to identify surveillance hovering over sensitive facilities.
- **GPS Spoofing & Spatial Teleportation:** Flags physically impossible spatial jumps (> 50 m/s delta) or sudden HDOP degradation indicating electronic GPS spoofing.
- **Power Collapse & Free-Fall Verification:** Correlates sudden battery voltage drops (< 10.5V) with high vertical acceleration spikes to pinpoint crash impact timestamps.
- **Geofence Breach Ray-Casting:** Executes point-in-polygon ray-casting against air-gapped geospatial geofence polygons, flagging unauthorized airspace incursions.

4.3 Raw Hex Offset Traceability & PDF Exhibit Generation

Garud FX provides a 3-column raw hexadecimal inspector ([Offset (8 Hex)] | [16 Hex Bytes] | [ASCII]). Every parsed telemetry coordinate links directly to its raw byte offset (e.g. Offset 0x0001F820). During courtroom cross-examination, examiners query the compiled .garud_db SQLite database container for instant spatial-temporal indexing and click any 2D/3D coordinate to jump directly to the exact raw byte offset in the binary dump.
Complete cases are exported into courtroom-ready PDF exhibits containing Executive Summaries, Flight Path Maps, Anomaly Logs, Hash-Chained Custody Manifests, and pre-filled Section 63(4) BSA Certificates.

5. DEVELOPMENT PLAN

The development plan for Stage 2 focuses on prototype enhancement, hardware drivers, and formal validation:

Milestone	Subsystem Focus	Technical Scope & Deliverables	Timeline	Success Metric
MS-01	Hardware Write-Blocker & Drivers	Integrate custom USB write-blocker firmware and direct SPI block	Weeks 1 - 2	100% write-block isolation verified via

		drivers for eMMC extractions.		hardware bus analyzer.
MS-02	Advanced Carving & Decryption	AES-128 decryption pipeline using pre-extracted firmware keys for DJI V4/V5 .DAT logs and corrupt header salvage.	Weeks 3 - 4	> 95% telemetry recovery on intentionally truncated crash logs.
MS-03	Real-Time Telemetry & Swarm Overlay	Implement live MAVLink stream decoding and multi-drone spatial trajectory alignment.	Weeks 5 - 6	Real-time rendering of 5 concurrent drone streams with < 50ms latency.
MS-04	Field-Deployable Hardware Kit	Build self-contained Linux ISO pre-configured with offline MBTiles vector maps for portable tablets.	Week 7	Complete air-gapped field operation on battery-powered rugged tablet.
MS-05	Formal CFTT & Legal Verification	Adapting NIST CFTT methodology to a self-constructed UAV reference dataset, red-team penetration, and mock courtroom cross-examination.	Week 8	100% pass rate on reference validation test cases and BSA certificate audits.

6. VALIDATION & TESTING METHODOLOGY

6.1 Test Environment & Repeatability Framework

Garud FX is validated within isolated, air-gapped forensic cleanrooms. The testing environment comprises Windows 11 Enterprise and Ubuntu 24.04 LTS workstations with hardware USB write-blockers, FTDI serial adapters, and SITL emulators running PX4 and ArduPilot flight dynamics. Validation utilizes reference raw images, field test crash logs, and synthetic anomaly injections following adapted NIST Computer Forensic Tool Testing (CFTT) principles for UAV platforms. Repeatability is confirmed by executing independent ingest runs on identical evidence images and verifying 100% identical SHA-256 output digests. If any evidence file or ledger block exhibits tampering, Garud FX halts ingestion, raises visual tamper alerts, and flags the compromised node.

6.2 Validation Test-Case Suite Results

Test ID	Test Description	Input Evidence	Expected Result	Observed Result	Status
TC-01	Physical Bit-Stream Hashing	eMMC_Physical.dd	Dual SHA-256 match	Dual SHA-256 generated matching raw bit-stream	**PASS**

TC-02	Magic-Byte Carving on Corrupted Logs	PX4_Crash_Corrupted.ulg	98.4% telemetry recovery	98.4% telemetry salvaged up to corruption horizon	**PASS**
TC-03	GCS vs. Onboard Timeline Correlation	Mavic3_Controller.txt & Onboard_DAT	Synchronized timeline	Clock drift adjusted; spatial delta < 0.5m	**PASS**
TC-04	Cryptographic Custody Chain Verification	custody.ledger (modified byte)	Instant tamper alarm	Ledger audit caught signature mismatch & halted ingest	**PASS**
TC-05	MAVLink Hardware Micro-Chunk USB Acquisition	Serial Connection (Pixhawk 4)	Replica .bin image	Read-only stream verified; zero write-back to FC flash	**PASS**
TC-06	Geofence Spatial Breach Ray-Casting	ArduPilot_Mission.bin	Automated breach alert	Point-in-polygon ray-casting flagged exact timestamp & offset	**PASS**
TC-07	Betaflight Blackbox High-Freq Decoding	Betaflight_Quad.bbl	1kHz Gyro & PID stream	100% frame synchronization with 0 byte offset loss	**PASS**
TC-08	Autel Explorer Logical Telemetry Extraction	Autel_Flight_Log.csv	Extracted flight parameters	Parsed 1,420 waypoints with full hex provenance mapping	**PASS**
TC-09	Yuneec DataPilot CSV Telemetry Parsing	Yuneec_Flight_Log.csv	Valid geodetic track & mode mapping	Successfully parsed timestamp, lat/lon, and mode state; 100% integrity	**PASS**